- Handle special cases like numerical data, dates, and named entities to preserve essential information.
- Validate the preprocessing pipeline to ensure no loss of critical data.

2. Extractive Summarization:
    - Implement extractive techniques that identify the most significant sentences from the text.
    - Use algorithms such as:
        - **TextRank:** A graph-based algorithm that ranks sentences based on their importance and interconnectivity.
        - **TF-IDF (Term Frequency-Inverse Document Frequency):** Scores sentences based on the uniqueness and relevance of their words.
    - Combine heuristics like sentence position, keyword density, and topic coverage for better results.
    - Visualize sentence importance scores to assist in fine-tuning the extractive process.

3. Abstractive Summarization:
    - Employ transformer-based models for generating summaries that paraphrase and restructure content:
        - Use models like **BART** (Bidirectional and AutoRegressive Transformers) or **GPT** (Generative Pre-trained Transformer).
        - Fine-tune these models on summarization datasets like CNN/Daily Mail or Gigaword to adapt them to specific use cases.
        - Balance between conciseness and informativeness by adjusting model parameters like length penalties.
    - Incorporate reinforcement learning techniques for improved abstractive summaries.